Chengle Fan

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Education

Southern University of Science and Technology — Shenzhen, China
B.Eng. in Optoelectronic Information Science and Engineering — *Expected June 2027*

- GPA: 3.85/4.00; Weighted Average: 92.45/100; Major Rank: 7/46.
- Selected coursework: Fundamentals of Optics (98/100), Engineering Electromagnetics (90/100), Solid-State Electronics (97/100), Introduction to Semiconductor Optics (92/100), Liquid Crystal Optoelectronics (95/100), Introduction to Wide Bandgap Semiconductors (96/100).

Nanyang Technological University — Singapore
Visiting Research Student (Non-Graduating) — *Jan. 12–Apr. 27, 2026*

Research Interests

Nanophotonics, integrated photonics, and optical computing. My interests span controlling optical modes and light–matter interactions in reconfigurable nanophotonic structures for tunable devices and sensing, as well as developing programmable photonic circuits for optical information processing and optical neural networks for artificial intelligence.

Manuscripts and Preprints

Chengle Fan[*], Kaizhou Shi[*], Xiang Xi[†], and Zhen Gao[†], "Observation of pseudospin-locked chiral edge transport in gyromagnetic photonic crystals", Submitted to *Physical Review Letters*. arXiv:2611.04827.

Shuxin Lin[*], Dong Zhao[*], **Chengle Fan**[*], Bei Yan, Yan Meng[†], Baile Zhang[†], and Zhen Gao[†], "Cavity-Tunable Dirac Phases in a Microwave Metasurface", Manuscript in preparation; intended submission to *Nature Physics*.

Minghui Niu[*], Yan Meng[*†], **Chengle Fan**, Xiaoyuan Jiao, Jidong Song, Perry Ping Shum, Zhen Gao[†], and Mengyao Li[†], "Cavity-tunable Pseudospin-Hall Topology and Edge Transport", Submitted to *Physical Review Letters*. arXiv:2610.20343.

[*]Equal contribution (co-first authors); [†]Corresponding authors.

Research Experience

Southern University of Science and Technology — Feb. 2025–Present
Undergraduate Research Assistant — *Supervisor: Prof. Zhen Gao*
Artificial Microstructure Physics Research Group

Pseudospin-Locked Chiral Edge States and Photonic Routing

- Led full-wave modeling and sample design of a gyromagnetic photonic crystal; characterized its topological phases using band inversion, pseudospin-resolved eigenfields, and spin Chern numbers.
- Designed measurement configurations and participated in most VNA transmission and near-field experiments, demonstrating approximately 20 dB directional transmission contrast and pseudospin-selective edge-mode excitation within the bandgap.
- Designed a pseudospin multiplexer/demultiplexer and a low-crosstalk waveguide crossing; wrote the first-authored manuscript and prepared its figures.

Cavity-Controlled Photonic Systems

- Contributed to sublattice-resolved VNA measurements of cavity-tunable Dirac phases, supporting pseudospin-texture reconstruction and winding-number characterization; co-first author on the manuscript.
- Investigate cavity-controlled refraction through MATLAB analysis of simulated bands, isofrequency contours, and interface phase matching, identifying candidate operating conditions for reconfigurable

beam routing.

- Contributed to experiments on cavity-controlled helical edge transport and to revision measurements for a study of cavity-tunable exceptional points.

Exploratory Modeling of Gauge-Field Photonic Lattices

- Developed tight-binding and full-wave models to explore gauge-field-induced duality and Hofstadter-type spectra in photonic lattices.

Nanyang Technological University Jan. 2026–Apr. 2026
Visiting Research Student *Supervisor: Prof. Baile Zhang*

Terahertz Chiral Valley Edge States

- Developed 2D and 3D COMSOL models of terahertz Chern–valley interfaces to evaluate overlapping bulk bandgaps and interface-localized modes.
- Optimized a magneto-optic slab geometry while assessing magnetic bias, fabrication tolerances, and finite air-layer effects.

Research Grant

SUSTech Undergraduate Research Training Program Shenzhen, China
Project Leader *June 2026–June 2027*

- Secured a competitive university research grant of CNY 10,000 for "Design of 3D Photonic Topological Insulators at Telecommunication Wavelengths."

Technical Skills

- **Computational electromagnetics:** COMSOL Multiphysics and FDTD in MATLAB.
- **Band topology and numerical modeling:** Tight-binding modeling and topological-invariant calculations.
- **Time-varying and spatiotemporal photonics:** MATLAB-based PWE/TMM band calculations, eigenfield reconstruction, and FDTD simulations for photonic time crystals and space–time-periodic media.
- **Experimental characterization:** VNA transmission and near-field mapping; optical alignment, CCD imaging, polarization microscopy, liquid-crystal cell assembly, and electro-optic characterization.
- **Programming and software:** MATLAB and COMSOL LiveLink; basic experience with C and Meep FDTD; familiarity with Silvaco, Zemax and TracePro.
- **Languages:** Mandarin Chinese (native); English (TOEFL iBT 104).

Honors and Awards

- SUSTech Outstanding Student Leader 2025
- SUSTech Outstanding Student 2024, 2025
- SUSTech Excellence Scholarship, Second Class / Third Class 2024, 2025

Leadership and Activities

- **Deputy Head, Activities Department**, EEE Student Union, SUSTech 2024–2025
- **Vice President**, SUSTech Orienteering Club 2024–2025